

Printer interface introduction

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1 Overview

1 PrinterLibs is jar bank to control printer under Android platform.

2 PrinterLibs has several public categories as below

A IO

It includes IO、BLEPrinting、BTPrinting、NETPrinting、USBPrinting

It realizes the basic Open、Close、Write、Read functions

IOCallback offers Open and Close return interfaces, and easy to get the connection status.

B Printing

It includes Pos、Label1

Pos realizes some functions like printing text, printing barcode, printing QR code and printing photos.

Label1 realizes label printing function, and need label printer to support.

C Use method

Android code fragment:

```
Pos mPos = new Pos();
BTPrinting mBt = new BTPrinting();
mPos.Set(mBt);
mBt.SetCallBack(this);
```

Then start asynchronous call:

```
public static class TaskOpen implements Runnable
{
    BTPrinting bt = null;
    String address = null;

    public TaskOpen(BTPrinting bt, String address)
    {
        this.bt = bt;
        this.address = address;
    }

    @Override
    public void run() {
        // TODO Auto-generated method stub
        bt.Open(address);
    }
}
```

```
}
```

Call the corresponding function to print when it connects successfully.

2 Interface

IO

Write

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data from the assigned excursion

count

The transmitted bytes

Return value

If it was written successfully, it returns the right bytes. Otherwise it returns -1

Remarks

The Write function of IO categories is empty, it always returns -1.

Read

Syntax

```
public int Read(byte[] buffer, int offset, int count, int timeout)
```

Parameters

buffer

Receive buffer area

offset

Store the received data to start from assigned excursion

count

Receiving bytes

Timeout

Exceeded time

Return value

If it was written successfully, it returns the right bytes. Otherwise it returns -1

Remarks

The Write function of IO categories is empty, it always returns -1.

IsOpened

Syntax

```
public boolean IsOpened()
```

Parameters

Return value

It returns TRUE if it connects printer. Or it returns FALSE

Remarks

The Write function of IO categories is empty, it always returns -1.

IOCallback

Deal with the four connection categories in the bottom:

BLEPrinting BTPrinting NETPrinting USBPrinting

When Open succeeds, it calls OnOpen

When Open fails, it calls OnOpenFailed

When Close or interrupt abnormally, it calls OnClose

OnOpen

It calls OnOpen when it connects successfully

Syntax

```
void OnOpen()
```

Parameters

Return value

Remarks

OnOpenFailed

It calls OnOpenFailed when it connects unsuccessfully

Syntax

```
void OnOpenFailed()
```

Parameters

Return value

Remarks

OnClose

It calls OnClose, when it breaks the connection(break automatically or interrupt abnormally).

Syntax

```
void OnClose()
```

Parameters

Return value

Remarks

BTPrinting

Bluetooth 2.0, read write control

Open

Connect assigned Bluetooth printer

Syntax

```
public boolean Open(String BTAddress)
```

Parameters

BTAddress

Bluetooth printer address: such as 00:11:22:33:44:55

Return value

It returns TRUE if it connects successfully, or it returns FALSE

It will recall interface OnOpen when it connects successfully, or it recalls OnOpenFailed

Listen

Connect 2.0 Bluetooth printer(it waits printer to connect automatically as main model)

Syntax

```
public boolean Listen(String BTAddress, int timeout)
```

Parameters

BTAddress

Bluetooth printer address: such as 00:11:22:33:44:55, it doesn't use mometarily

timeout

Wait the exceeded time

Return value

It returns TRUE if it connects successfully, or it returns FALSE

Remarks

It will recall interface OnOpen when it connects successfully, or it recalls OnOpenFailed

Close

Close connection

Syntax

```
public void Close()
```

Parameters

Return value

Remarks

It closes the connection, and recalls interface OnClose, it won't recall many time if you repeat close.

Write

Write data via Bluetooth

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data start from the assigned excursion

count

The transmitting bytes

Return value

If it is written successfully, it returns the right bytes. Or it returns -1.

Remarks

Read

Read data

Syntax

```
public int Read(byte[] buffer, int offset, int count, int timeout)
```

Parameters

buffer

Receive buffer area

offset

Store received data to start from assigned excursion

count

Receiving bytes

timeout

Exceeded time

Return value

If it is written successfully, it returns the right bytes. Or it returns -1.

Remarks

SkipAvailable

Ignore the data in buffer area

Syntax

```
public void SkipAvailable()
```

Parameters

Return value

Remarks

IsOpened

If it is connected?

Syntax

```
public boolean IsOpened()
```

Parameters

Return value

It returns TRUE to mean it connects. Or it returns FALSE

Remarks

IsOpened function is built base on heart beats, and cannot get real connection status.

If printer is off accidently, IsOpened may take a few seconds to return right result.

You can use RTQueryStatus of POS system function if you want to make sure if the printer is connected.

SetCallback

Set recall interface

Syntax

```
public void SetCallback(IOCallBack callBack)
```

Parameters

callBack

Recall interface only if set this when connect it successfully or not.

Return value

Remarks

BLEPrinting

Bluetooth 4.0, read write control

Open

Connect assigned Bluetooth printer

Syntax

```
public boolean Open(String BTAddress)
```

Parameters

BTAddress

Bluetooth printer address: such as 00:11:22:33:44:55

Return value

It returns TRUE if it connects successfully. Or it returns FALSE.

Remarks

It recall interface OnOpen when it connects successfully, or it recalls OnOpenFailed

Close

Close connection

Syntax

```
public void Close()
```

Parameters

Return value

Remarks

Close connection to recall interface OnClose, it won't recall many times if repeats close.

Write

Write data via Bluetooth

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data to start from assigned excursion

count

Receiving bytes

Return value

It returns the right bytes if it is written successfully, or it returns -1.

Remarks

The printing speed of Bluetooth 4.0 is much slower than 2.0 version

Read

Read data

Syntax

```
public int Read(byte[] buffer, int offset, int count, int timeout)
```

Parameters

buffer

Receive buffer area

offset

It stores the received data to start from assigned excursion

count

Receiving bytes

timeout

Exceeding time

Return value

It returns the right bytes if it is written successfully, or it returns -1.

Remarks

SkipAvailable

Ignore data in buffer area

Syntax

```
public void SkipAvailable()
```

Parameters

Return value

Remarks

IsOpened

If it is connected?

Syntax

```
public boolean IsOpened()
```

Parameters

Return value

It returns TRUE to mean it connects, or it returns FALSE.

Remarks

IsOpened may time out.

You can use RTQueryStatus in POS series function if you want to make sure if the printer is connected.

SetCallback

Set callback interface

Syntax

```
public void SetCallback(IOWallBack callBack)
```

Parameters

callBack

It has Callback only to set this command when it connect ssuccessfully
or not for callback interface

Return value

Remarks

NETPrinting

WIFI bottom connection, read write control

Open

Connect assigned internet printer

Syntax

```
public boolean Open(String IPAddress, int PortNumber)
```

Parameters

IPAddress

Printer IP address: it can get in printer self-test page, printer 打印
default IP: 192.168.1.87

PortNumber

Printer port no.: fixed is 9100

Return value

It returns TRUE if it connects successfully, or it returns FALSE.

Remarks

It calls back interface OnOpen when it connects successfully, or it calls back OnOpenFailed

Close

Close connection

Syntax

```
public void Close()
```

Parameters

Return value

Remarks

Close connection to call back interface OnClose, it won't call back many time if it repeats Close.

Write

Write data via internet port

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data to start from assigned excursion

count

Transmitting bytes

Return value

It returns the right bytes if it is written successfully, or it returns -1.

Remarks

It may cause block if WIFI signal is bad or internet is not good. Or there won't be a problem to print under normal situation.

Read

Read data

Syntax

```
public int Read(byte[] buffer, int offset, int count, int timeout)
```

Parameters

buffer

Receive buffer area

offset

It stores data to start from assigned excursion

count

Receiving bytes

timeout

Exceeding time

Return value

It returns the right bytes if it is written successfully, or it returns -1.

Remarks

SkipAvailable

Ignore data in buffer area

Syntax

```
public void SkipAvailable()
```

Parameters

Return value

Remarks

IsOpened

If it is connected?

Syntax

```
public boolean IsOpened()
```

Parameters

Return value

It returns TRUE to show it is connected. Or it returns FALSE.

Remarks

IsOpened function is built based on heart beat, and cannot get real-time connection status.

If the printer is off accidentally, IsOpened may take a few seconds to return right result.

If you want to check if the printer is connected, you can use RTQueryStatus in POS series function.

SetCallBack

Set recall interface

Syntax

```
public void SetCallBack(IOWallBack callBack)
```

Parameters

callBack

Recall interface only if set this when connect it successfully or not.

Return value

Remarks

USBPrinting

USB bottom connection, read write control

Open

Connect assigned USB printer

Syntax

```
public boolean Open(UsbManager manager, UsbDevice device)
```

Parameters

manager

UsbManager

Use (UsbManager) getSystemService(Context.[USB_SERVICE](#)) to get

device

UsbDevice

Get UsbDevice (mUsbManager.getDeviceList()) via USB device

Return value

It returns TRUE if it connects successfully, or it returns FALSE.

Remarks

It calls back interface OnOpen when it connects successfully, or it calls back OnOpenFailed

Close

Close connection

Syntax

```
public void Close()
```

Parameters

Return value

Remarks

Close connection, it calls back interface OnClose, it won't call back many time if repeat Close

Write

Via USB to write data

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data to start from assigned excursion

count

Receiving bytes

Return value

If it is written successfully, it returns right bytes, or it returns -1.

Remarks

USB transmitting data speed is the quickest and most stable, we suggest to use USB to print.

Read

Read data

Syntax

```
public int Read(byte[] buffer, int offset, int count, int timeout)
```

Parameters

buffer

Receive buffer area

offset

Store data to start from assigned excursion

count

Receiving byte

timeout

Exceeding time

Return value

If it reads successfully, it returns the right bytes. Or it returns -1.

Remarks

SkipAvailable

Ignore data in buffer area

Syntax

```
public void SkipAvailable()
```

Parameters

Return value

Remarks

IsOpened

If it is connected?

Syntax

```
public boolean IsOpened()
```

Parameters

Return value

It returns TRUE if it connects. Or it returns FALSE.

Remarks

IsOpened function is built based on heart beat, and can't get real-time online status.

If printer is off accidentally, IsOpened may take a few seconds to return right result.

If you want to make sure if printer connects, you can use RTQeuryStatus in POS function.

SetCallBack

Set recall interface

Syntax

```
public void SetCallBack(IOCallBack callBack)
```

Parameters

callBack

Recall interface only if set this when connect it successfully or not.

Return value

Remarks

MemoryPrinting

USB bottom connection, read write control

Write

rite data to memory

Syntax

```
public int Write(byte[] buffer, int offset, int count)
```

Parameters

buffer

Transmit buffer area

offset

Send data to start from assigned excursion

count

Receiving bytes

Return value

If it is written successfully, it returns right bytes, or it returns -1.

Remarks

GetWriteBuffer

Get Memory Buffer

Syntax

```
public byte[] GetWriteBuffer()
```

Parameters

Return value

return memory buffer and clear it.

Remarks

Pos

POS communicate with printer via a IO object
Use Set(IO) to set POS IO object
The subsequent commands are transmitted via assigned IO

Set

Assign IO object

Syntax

```
public void Set(IO io)
```

Parameters

io

Need to use IO object

Return value

Remarks

Call this function to bind the bottom read and write object to the top logic object.

GetIO

Get the current IO object

Syntax

```
public IO GetIO() ()
```

Parameters

Return value

Current IO object

Remarks

POS_PrintPicture

Print picture

Syntax

```
public void POS_PrintPicture(Bitmap mBitmap, int nWidth, int  
nBinaryAlgorithm, int nCompressMethod)
```

Parameters

mBitmap

Printing bitmap

nWidth

Printing width

If nWidth and Bitmap width is different, it waits the percentage to turn nWidthw width

The max. width doesn't exceed 384 dots of 2 inches printer(58mm printer)

The max. width doesn't exceed 576 dots of 3 inches printer(80mm printer)3

nBinaryAlgorithm

Two-valued conversion method

0 uses shake method, which has good efficiency to colorful pictures.

1 uses average threshold value method, which has good efficiency to text pictures.

nCompressMethod

Compress method

0 doesn't use compress method

1 uses compress method

Return value

Remarks

Print picture to use shake method, which has good efficiency to colorful picture printing. And has gray level efficiency.

POS_S_TextOut

Print character set as specified format

Syntax

```
public void POS_S_TextOut (String pszString, int nOrgx, int nWidthTimes,  
                           int nHeightTimes, int nFontType, int nFontStyle)
```

Parameters

pszString

Need to print character set

nOrgx

Assign dots from the start point position to left margin in x direction(horizontal)

2 inches printer is 384 dots/line, 3 inches printer is 576 dots/line.

nWidthTimes

Assign magnification times of character width direction.

It can be 0 to 1.

nHeightTimes

Assign magnification time of character length direction.

It can be 0 to 1.

nFontType

Assign character font type.

(0x00 Standard ASCII 12x24)

(0x01 Constringent ASCII 9x17)

nFontStyle

Assign character font style. It can be one of some styles as below.

(0x00 Normal)

(0x08 Bold)

(0x80 1 dot bold underline)

(0x100 2 dots bold underline)

(0x200 convert, only efficient in line start)

(0x400 inverse, white in black)

(0x1000 every character rotates clockwise 90 °)

Return value

Remarks

POS_TextOut

Print character set as specified format (can select language)

Syntax

```
public void POS_S_TextOut(String pszString, int nLan, int nOrgx, int nWidthTimes, int nHeightTimes, int nFontType, int nFontStyle)
```

Parameters

pszString

Need to print character set

nLan

0-GBK 1-UTF8 3-BIG5 4-SHIFT-JIS 5-EUC-KR

nOrgx

Assign dots from the start point position to left margin in x direction(horizontal)

2 inches printer is 384 dots/line, 3 inches printer is 576 dots/line.

nWidthTimes

Assign magnification times of character width direction.

It can be 0 to 1.

nHeightTimes

Assign magnification time of character length direction.

It can be 0 to 1.

nFontType

Assign character font type.

(0x00 Standard ASCII 12x24)

(0x01 Constringent ASCII 9x17)

nFontStyle

Assign character font style. It can be one of some styles as below.

(0x00 Normal)

(0x08 Bold)

(0x80 1 dot bold underline)

(0x100 2 dots bold underline)

(0x200 convert, only efficient in line start)

(0x400 inverse, white in black)

(0x1000 every character rotates clockwise 90 °)

Return value

Remarks

POS_S_SetBarcode

Print barcode

Syntax

```
public void POS_S_SetBarcode(String strCodedata, int nOrgx, int nType,
                             int nWidthX, int nHeight, int nHriFontType,
                             int nHriFontPosition)
```

Parameters

strCodedata

It needs the barcode character sets

Some barcodes has format requirements, pls print barcodes as rules.

nOrgx

It assigns dots from start position to left margin in x direction(horizontal).

2 inches printer is 384 dots/line, 3 inches printer is 576 dots/line

nType

Assign barcode type

It can be any value as below chart.

Value	Meaning
0x41	UPC-A
0x42	UPC-C
0x43	JAN13(EAN13)
0x44	JAN8(EAN8)
0x45	CODE39
0x46	ITF
0x47	CODEBAR
0x48	CODE93
0x49	CODE 128

nWidthX

Assign barcode basic element width

Range:[2,6]

nHeight

Assign barcode height dots

It can be from 1 to 255

`nHriFontType`

Assign HRI (Human Readable Interpretation) character font type.

It can be any value as below chart

Value	Meaning
0x00	Standard ASCII
0x01	Constringent ASCII

`nHriFontPosition`

Assign HRI (Human Readable Interpretation) character position

It can be any value as below chart

Value	Meaning
0x00	Doesn't print
0x01	Only print in barcode upward
0x02	Only print in barcode downward
0x03	Print both barcode upward and downward

Return value

Remarks

The barcode won't be printed if barcode is too wide to print, which exceeds the max. printing width.

Barcode won't be printed if there is something wrong with barcode format

POS_S_SetQRcode

Print QR code

Syntax

```
public void POS_S_SetQRcode(String strCodedata, int nWidthX,  
int nVersion, int nErrorCorrectionLevel)
```

Parameters

strCodedata

QR code character string

nWidthX

Unit width of each module for QR code, [1,16]

Set module width properly can make QR code nicer

nVersion

QR code version size, this value is about QR code size. [0,16]

Set as 0 to calculate QR code version size automatically

Pls set proper value for this value if you want the QR code size to be fixed.

nErrorCorrectionLevel

Error correction level

[1,4]

Return value

Remarks

POS_DoubleQRcode

Print Double QR code in one line

Syntax

```
public void POS_DoubleQRCode(  
    String QR1Data, int QR1Position, int QR1Ecc, int QR1Version,  
    String QR2Data, int QR2Position, int QR2Ecc, int QR2Version,  
    int ModuleSize)
```

Parameters

QR1Data

QR1 data

QR1Position

QR1 Position (2inch printer max width 384, 3 inch max 576)

QR1Ecc

QR1 error correction level (1-4)

QR1Version

QR1 version (0-40, 0 auto select)

QR2Data

QR2 data

QR2Position

QR2 Position

QR2Ecc

QR2 error correction level (1-4)

QR2Version

QR2 version (0-40, 0 auto select)

ModuleSize

QR Code Module Size

Return value

Remarks

POS_FeedLine

Feed paper one line

Syntax

```
public void POS_FeedLine()
```

Parameters

Return value

Remarks

POS_S_Align

Set align

Syntax

```
public void POS_S_Align(int align)
```

Parameters

Align

Set align

(0 align left)

(1 align center)

(2 align right)

Return value

Remarks

POS_SetLineHeight

Set line height

Syntax

```
public void POS_SetLineHeight(int nHeight)
```

Parameters

nHeight

Line height (0,255]

Return value

Remarks

POS_Reset

`Reset printer (software resets)`

Syntax

`public void POS_Reset ()`

Parameters

Return value

Remarks

POS_SetMotionUnit

Set printer motion unit

Syntax

```
public void POS_SetMotionUnit(int nHorizontalMU, int nVerticalMU)
```

Parameters

nHorizontalMU

Set motion unit as 25.4/ nHorizontalMU mm in horizontal direction

nVerticalMU

Set motion unit as 25.4 / nVerticalMU mm in vertical direction.

Return value

Remarks

POS_S_SetAreaWidth

Set printing area width under standard model

Syntax

```
public void POS_S_SetAreaWidth(int nWidth)
```

Parameters

nWidth

Assign printing area width

Return value

Remarks

POS_CutPaper

Cut paper

Syntax

```
public void POS_CutPaper ()
```

Parameters

Return value

Remarks

It is only efficient for machine with cutter

POS_Beep

Buzzer beeps

Syntax

```
public void POS_Beep(int nBeepCount, int nBeepMillis)
```

Parameters

nBeepCount

Beep times

nBeepMillis

Beep time each time = 100 * nBeepMillis ms

Return value

Remarks

POS_KickDrawer

Open cashbox

Syntax

```
public void POS_KickDrawer(int nDrawerIndex, int nPulseTime)
```

Parameters

nDrawerIndex

0 means: pulse was sent to cashbox to output pin 2

1 means: pulse was sent to cashbox to output pin 5

nPulseTime

Pulse time

High electric level: $nPulseTime * 2ms$

Low electric level : $nPulseTime * 2ms$

Return value

Remarks

POS_SetPrintSpeed

Set print speed remark: if print speed is faster than transmitting speed, printer exists block.

Syntax

```
public void POS_SetPrintSpeed(int nSpeed)
```

Parameters

nSpeed

Print speed (mm/s)

Return value

Remarks

Set print speed as data to send speed, which can reach the best print efficiency.
It measures length and time via printing a receipt. And use length/time to show.

POS_QueryStatus

Query status

This command is blocked all the time when printer is busy.

The return status stores in status

Syntax

```
public boolean POS_QueryStatus(byte[] status, int timeout, int MaxRetry)
```

Parameters

status

status = new byte[1] this value has no meaning so far

timeout

Single time to query status exceeds time

MaxRetry

Fail retry times

Return value

It returns TRUE to show printer status is OK. Or printer is not connected or printer is busy.

Remarks

POS_RTQueryStatus

Real-time query status

No matter under any status of the printer, it returns the status when it receives this command.

The return status stores in status

Syntax

```
public boolean POS_RTQueryStatus(byte[] status, int type, int timeout,
    int MaxRetry)
```

Parameters

status

```
status = new byte[1]
```

type

```
type can be [1,4]
```

1: Printer status

Bit	0/1	Hex.	Dec.	Function
0	0	00	0	Fixed is 0
1	1	02	2	Fixed is 1
2	0	00	0	One or two cashboxes are open(Fixed is 0 if this machine is without cashbox)
	1	04	4	Two cashboxes are closed
3	0	00	0	Online
	1	08	8	Offline
4	1	10	16	Fixed is 1
5, 6		--	--	Undefined
7	0	00	00	Paper has been tore off
	1	80	96	Paper hasn't been tore off

2: Transmit offline status

Bit	0/1	Hex.	Dec.	Function
0	0	00	0	Fixed is 0
1	1	02	2	Fixed is 1
2	0	00	0	Upper cover is closed

	1	04	4	Upper cover is open
3	0	00	0	Paper feed key is un-pressed
	1	08	8	Paper feed key is pressed
4	1	10	16	Fixed is 1
5	0	00	0	Paper is not out
	1	20	32	Paper out
6	0	00	00	No error
	1	40	64	Error
7	0	00	0	Fixed is 0

3: Transmit error status

Bit	0/1	Hex.	Dec.	Function
0	0	00	0	Fixed is 0
1	1	02	2	Fixed is 1
2		--	--	Undefined
3	0	00	0	Cutter has no error
	1	08	8	Cutter has error
4	1	10	16	Fixed is 1
5	0	00	0	No recoverable error
	1	20	32	Has recoverable error
6	0	00	00	Printer head temp. and voltage are normal
	1	40	64	Printer head temp. or voltage is out of range
7	0	00	0	Fixed is 0

4: Transmit paper sensor status

Bit	0/1	Hex.	Dec	Function
0	0	00	0	Fixed is 0
1	1	02	2	Fixed is 1
2, 3	0	00	0	With paper
	1	0C	12	Paper will be out
4	1	10	16	Fixed is 1
5, 6	0	00	0	With paper
	1	60	96	Paper out
7	0	00	0	Fixed is 0

timeout

timeout ms

MaxRetry

Fail retry times

Return value

It returns TRUE to show printer communication is normal. The query status stores in status.

Remarks

POS_TicketSucceed

Transmit data query command

Syntax

```
public boolean POS_TicketSucceed(int dwSendIndex, int timeout)
```

Parameters

dwSendIndex

Receipt index:

It can increase accordingly from 1, and has no meaning so far.

timeout

The exceeding ms time for waiting receipt to print

Return value

Receipt printing is finished, and has no reason to interrupt for paper out, it returns TRUE.

Otherwise it doesn't check the status, or returns printing interruption as paper out or other errors, then it returns FALSE.

Remarks

To make sure the reliability of the receipt printing. Pls call this function to ensure the receipt printing result every time when you finish the printing task.

Label

Label communicate with printer via a IO object

Use `Set(IO)` to set Label IO object

The subsequent commands are transmitted via assigned IO

Set

Assign IO object

Syntax

```
public void Set(IO io)
```

Parameters

io

Need to use IO object

Return value

Remarks

Call this function to bind the bottom read and write object to the top logic object.

GetIO

Get the current IO object

Syntax

```
public IO GetIO() ()
```

Parameters

Return value

Current IO object

Remarks

PageBegin

Description: assign the start of a Page, and set Page size, reference point coordinates and page rotating angle.

Parameters

Syntax

```
public void PageBegin(int startx, int starty, int width, int height,  
int rotate)
```

Parameters

startx

page start point x coordinates

starty

page start point y coordinates

width

page width

startx + width range is [1,384]. The printer is 384 dots/line when editing SDK. Pls refer to printer specification if you are not sure the dots number per line. Normally there are three specification 384,576,832.

height

page height

starty + height range is [1,936]. The limited is 936 when editing SDK. But this value is not sure, which is about the printer resource. Even though, we don't suggest to set page width too wide. We suggest to set page width and height to be fit for label paper.

rotate

page rotating. The value range of rotate is {0,1}. Page doesn't rotate to print as 0, and rotate 90° to print as 1.

Return value

Remarks

PageEnd

Description: It is only the end of a Page

Syntax

```
public void PageEnd()
```

Parameters

Return value

Remarks

PagePrint

Description: print the Page contents to label paper

Syntax

```
public void PagePrint(int num)
```

Parameters

num

printing times,1-255.

Return value

Remarks

PageFeed

Feed paper to gap

Syntax

```
public void PageFeed()
```

Parameters

Return value

Remarks

DrawPlainText

Description: draw text in assigned position of Page.
only for single line

Syntax

```
public void DrawPlainText(int startx, int starty, int font, int style,
                          byte[] str)
```

Parameters

startx

define text start position x coordinates,the value range is: [0, Page_Width-1]

starty

define text start position y coordinates,the value range is: [0, Page_Height-1]

font

Choose font, the efficient value range is {16, 24, 32, 48, 64, 80, 96},this printer only can use 24.

style

chracter style.

Databits	define
0 Bold flag bit:	font bold for 1,don' t bold if reset zero clearing.
1 underline flag bit:	underline text for 1, don' t underline if rest zero clearing
2 inverse flag bit:	inverse for 1(white in black), don' t inverse rest zero clearing
3 delete line flage bit:	for 1 text with delete line,don' t delete line if reset zero clearing.
[5,4] rotate flag bit:	00 rotates 0°
	01 rotates 90°
	10 rotates 180°
	11 rotates 270°
[11,8] font width magnification times;	
[15,12] font height magnification times;	

`str`

character set data flow

Return value

Remarks

DrawLine

Description: Draw a straightway in the assigned position of Page

Syntax

```
public void DrawLine(int startx, int starty, int endx, int endy, int width,  
                    int color)
```

Parameters

startx

straightway start point x coordinates,the value range is: [0, Page_Width-1]

starty

straightway start point y coordinates,the value range is: [0, Page_Height-1]

endx

straightway end point x coordinates,the value range is: [0, Page_Width-1]

endy

straightway end point y coordinates,the value range is: [0, Page_Height-1]

width

straightway width,the value range is: [1, Page_Height-1]

color

straightway color,
the value range is: {0, 1}.
When Color is 1, straightway is black.
When Color is 0, straightway is white.

Return value

Remarks

DrawBox

Description: draw assigned size rectangular frame in the fixed position of Page

Syntax

```
public void DrawBox(int left, int top, int right, int bottom,  
                    int borderwidth, int bordercolor)
```

Parameters

left

rectangular frame top left corner x coordinates, the value range is:
[0, Page_Width-1]

top

rectangular frame top left corner y coordinates, the value range is:
[0, Page_Height-1]

right

rectangular frame top right corner x coordinates, the value range is:
[0, Page_Width-1]

bottom

rectangular frame top right corner y coordinates, the value range is:
[0, Page_Height-1]

borderwidth

rectangular line width

bordercolor

rectangular line color,
range is {0, 1}.

When Color = 1 ,draw black rectangular ,
when Color = 0 ,draw white rectangular

Return value

Remarks

DrawRectangel

Description: draw rectangular frame in assigned position of Page

Syntax

```
public void DrawRectangel(int left, int top, int right, int bottom,  
                           int color)
```

Parameters

left

rectangular block top left corner x coordinates, the value range is: [0, Page_Width-1]

top

rectangular block top left corner y coordinates, the value range is: [0, Page_Height-1]

right

rectangular block top right corner x coordinates, the value range is: [0, Page_Width-1]

bottom

rectangular block top right corner y coordinates, the value range is: [0, Page_Height-1]

color

rectangular block color,
the value range is: {0, 1}.

When color is 1, rectangular block is black.

When Color is 0, rectangular block is white.

Return value

Remarks

DrawBarcode

Description: Draw 1D code in the assigned position of Page

Syntax

```
public void DrawBarcode(int startx, int starty, int type, int height,  
                        int unitwidth, int rotate, byte[] str)
```

Parameters

startx

barcode top left corner x coordinates, the value range is: [0, Page_Width-1]

starty

barcode top left corner y coordinates, the value range is: [0, Page_Height-1]

type

indetify barcode type, the value range is: [0, 29]. Definitions are as below:

type	name	length	barcode value range (Dec.)
0	UPC-A	11	48-57
1	UPC-E	6	48-57
2	EAN13	12	48-57
3	EAN8	7	48-57
4	CODE39	1-	48-57, 65-90, 32, 36, 37, 43, 45, 46, 47
5	I25	1-even number	48-57
6	CODABAR	1-	48-57, 65-68, 36, 43, 45, 46, 47, 58
7	CODE93	1-255	0-127
8	CODE128	2-255	0-127
9	CODE11		
10	MSI		
11	128M		
12	EAN128		
13	25C		
14	39C		
15	39		
16	EAN13+2		
17	EAN13+5		

18	EAN8+2		
19	EAN8+5		
20	POST		
21	UPCA+2		
22	UPCA+5		
23	UPCE+2		
24	UPCE+5		
25	CPOST		
26	MSIC		
27	PLESSEY		
28	ITF14		
29	EAN14		

`height`

define barcode height

`unitwidth`

It assigns the basic element width.

It can be any value(n) in the chart as below

n	Single basic module width(succesive type)	Double basic module width(discrete type)	
		Narrow element width	Wide element width
2	0. 25mm	0. 25mm	0. 625mm
3	0. 375mm	0. 375mm	1. 0mm
4	0. 5mm	0. 5mm	1. 25mm
5	0. 625mm	0. 625mm	1. 625mm

`rotate`

Mean barcode rotating angle,

the value range is: [0, 3].

Definitions are as below:

Rotate value define

0 barcode doesn' t rotate to draw

1 barcode rotates 90° draw.

2 barcode rotates 180° draw.

3 barcode rotates 270° draw

`str`

Text character data flow

Return value

Remarks

DrawQRCode

Description: Draw QR code in assigned position of Page

Syntax

```
public void DrawQRCode(int startx, int starty, int version, int ecc,
                        int unitwidth, int rotate, byte[] str)
```

Parameters

startx

QRCode code top left corner x coordinates, the value range is: [0, Page_Width-1]

starty

QRCode code top left corner y coordinates, the value range is: [0, Page_Height-1]

version

Assign charater version. The value range is:[0,20]. When version is 0, printer caculates version number according to character set automatically.

ecc

Assign error correction level.

The value range is: [1, 4].

Definitios are as below:

ECC error correction level

1 L: 7%, low error correction, much data.

2 M: 15%, medium error correction

3 Q: optimize error correction

4 H: 30%, the highest error correction, less data.

unitwidth

QRCode code block, the value range Is:[1, 4]. Definitions are the same parameter as UniWidth of 1D code command.

rotate

QRCode code rotating angle, the value range is: [0, 3]. Definitions are the smae paramter as Rotate of 1D code command.

str

QRCode test character data flow

Return value

Remarks

DrawPDF417

Description: draw PDF417 code in the assigned position of Page

Syntax

```
public void DrawPDF417(int startx, int starty, int colnum, int lwratio,
                      int ecc, int unitwidth, int rotate, byte[] str)
```

Parameters

startx

PDF417 code top left corner x coordinates, the value range is: [0, Page_Width-1]

starty

PDF417 code top left corner y coordinates, the value range is: [0, Page_Height-1]

colnum

ColNum is colnum, which means how many digits in per line. A digit is 17*UnitWidth dots. Line number is produces automatically by printer, the limited range is 3~90. ColNum value range:[1,30].

lwratio

width height ratio. The value range: [3,5]

ecc

Error correction level, the value range is: [0. 8]

Ecc value error correction number stored files number(byte)

0 2 1108

1 4 1106

2 8 1101

3 16 1092

4 32 1072

5 64 1024

6 128 957

7 256 804

8 512 496

unitwidth

PDF417 code block, the value range is: [1, 3]. Definitions are the same parameter as UniWidth of 1D code.

`rotate`

PDF417 code rotating angle, the value range is: [0, 3]. Definitions are the parameter as Rotate of 1D code

`str`

PDF417 text character data flow.

Return value

Remarks

DrawBitmap

Description: Draw picture in the assigned position of Page

Syntax

```
public void DrawBitmap(int startx, int starty, int width, int height,
                        int style, byte[] pdata)
```

Parameters

startx

Bitmap top left corner x coordinates, the value range is: [0, Page_Width]

starty

Bitmap top left corner y coordinates, the value range is: [0, Page_Height]

width

Bitmap pixel width

height

Bitmap pixel height

style

Bitmap print special effects, and definitions are as below:

bit	definition
0	inverse flag bit, inverse 1 bitmap to print, and rest to print
[2:1]	rotates flag bit 00 Rotate 0° 01 rotates 90° 10 rotates 180° 11 rotates 270°
[7:3]	reserved
[11:8]	Bitmap width magnification times.
[15:12]	bitmap height magnification times

pdata

Bitmap dot matrix data

Return value

Remarks

DrawBitmap

Description: Draw picture in the assigned position of Page

Syntax

```
DrawBitmap(int startx, int starty, int dstWidth, int dstHeight, int style,  
Bitmap mBitmap, int nBinaryAlgorithm)
```

Parameters

startx

Bitmap top left corner x coordinates, the value range is: [0, Page_Width]

starty

Bitmap top left corner y coordinates, the value range is: [0, Page_Height]

dstWidth

the print width

dstHeight

the print height

style

same as above

mBitmap

the bitmap to print

nBinaryAlgorithm

Two-valued conversion method

0 uses shake method, which has good efficiency to colorful pictures.

1 uses average threshold value method, which has good efficiency to text pictures.

Return value

Remarks

Page

Page communicate with printer via a IO object
Use Set(IO) to set Page IO object
The subsequent commands are transmitted via assigned IO

Set

Assign IO object

Syntax

```
public void Set(IO io)
```

Parameters

io

Need to use IO object

Return value

Remarks

Call this function to bind the bottom read and write object to the top logic object.

GetIO

Get the current IO object

Syntax

```
public IO GetIO() ()
```

Parameters

Return value

Current IO object

Remarks

PageEnter

Choose page mode

Syntax

```
public boolean PageEnter()
```

Parameters

Return value

If the write command successfully, return TURE, or return FALSE

Remarks

PagePrint

Printing content at Page mode

Syntax

```
public boolean PagePrint()
```

Parameters

Return value

If the write command successfully, return TURE, or return FALSE

Remarks

PageExit

Exit page mode

Syntax

```
public boolean PageExit ()
```

Parameters

Return value

If the write command successfully, return TURE, or return FALSE

Remarks

SetPrintArea

Setting printing area at page mode

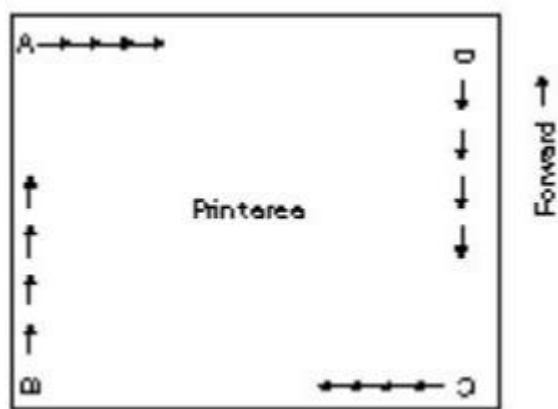
Syntax

```
public boolean SetPrintArea(int left, int top, int right, int bottom, int direction)
```

Parameters

- left
 - x coordinate of left corner printing area
- top
 - Y coordinate of top left corner printing area
- right
 - X coordinate of bottom right corner
- bottom
 - Y coordinate of bottom right corner
- direction
 - Direction of printing area

direction	Printing direction	Starting position
0	from left to right	Top left corner(A in picture)
1	from bottom to top	Bottom left corner (B in picture)
2	from right to left	Bottom right corner (C in picture)
3	from top to bottom	Top right corner (D in picture)

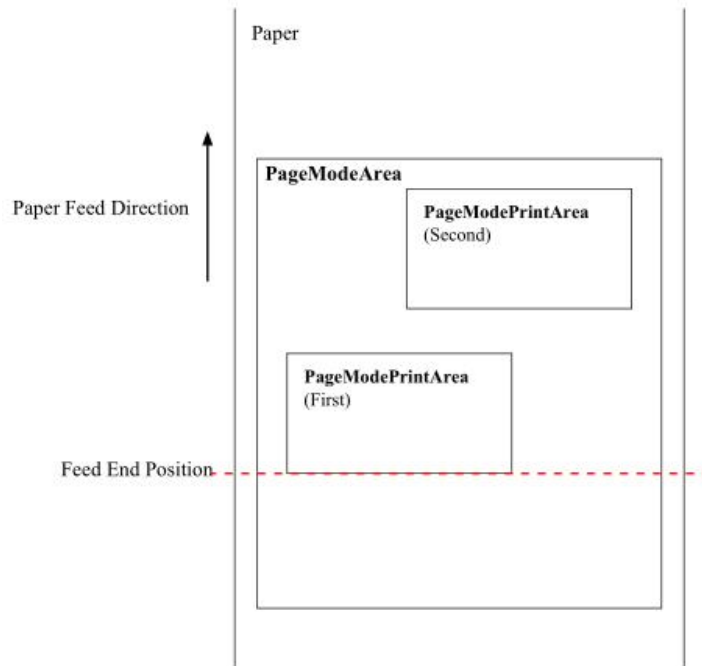


Return value

If the write command successfully, return TRUE, or return FALSE

Remarks

Following picture show the concept of printing area



DrawText

drawing the text

Syntax

```
public boolean DrawText(String pszString, int x, int y, int nWidthScale,  
int nHeightScale, int nFontType, int nFontStyle)
```

Parameters

pszString

content need to print。 UNICODE code strings

x

specify number of dots from starting area of horizontal direction to left margin of printing area. （horizontal axis）

support left center right alignment

incoming x see table

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

specify number of dots from starting area of vertical direction to upper edge of printing area.(vertical direction)

nWidthScale

specified width magnification [0,7]

nHeightScale

specified height magnification [0,7]

nFontType

font type

0 standard font

1 compressed font

nFontStyle

specify font styles, can be one or more of following table,(add together)

Value	Meaning
0x00	normal
0x08	Bold
0x80	1 dot bold underline
0x100	2 dot bold underline
0x200	invert (only valid at the head of first line)
0x400	reverse (black and white)
0x1000	rotate each character 90 degrees clockwise

Return value

If the write command successfully, return TRUE, or return FALSE

Remarks

Page mode Draw functions won't print at once, just fill in the page, till call PAGE_PagePrint , it will print.

DrawBarcode

draw barcode

Syntax

```
public boolean DrawBarcode(String pszString, int x, int y, int nBarcodeUnitWidth, int nBarcodeHeight, int nHriFontType, int nHriFontPosition, int nBarcodeType)
```

Parameters

pszString

Draw barcode

x

It assigns the dots from the start point position to printing area margin in horizontal direction.(Horizontal coordinates)

It supports left justifying, mediate, right justifying

Introduce x as below chart

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

It assigns the dots from start point position to printing area margin in vertical direction(horizontal coordinates)

nBarcodeUnitWidth

It assigns the basic element width.

It can be any value(n) in the chart as below

n	Single basic module width(succe ssive type)	Double basic module width(discrete type)	
		Narrow element width	Wide element width
2	0. 25mm	0. 25mm	0. 625mm
3	0. 375mm	0. 375mm	1. 0mm
4	0. 5mm	0. 5mm	1. 25mm
5	0. 625mm	0. 625mm	1. 625mm

6	0.75mm	0.75mm	1.875mm
---	--------	--------	---------

nBarcodeHeight

Barcode height

nHriFontType

It assigns HRI (Human Readable Interpretation) character font types.

It can be any value (n) of the chart as below.

Value	Meaning
0x00	standard ASCII
0x01	compressed ASCII

nHriFontPosition

It assigns HRI (Human Readable Interpretation) character position

It can be any value (n) of the chart as below.

Value	Meaning
0x00	Doesn't print
0x01	Only print in upward side of barcode
0x02	Only print in downward side of barcode
0x03	Print both in upward side and downward

nBarcodeType

It can be any value as below chart

Value	Meaning
0x41	UPC-A
0x42	UPC-C
0x43	JAN13(EAN13)
0x44	JAN8(EAN8)
0x45	CODE39
0x46	ITF
0x47	CODEBAR
0x48	CODE93
0x49	CODE 128

Return value

If the write command successfully, return TRUE, or return FALSE

Remarks

currently not support barcode alignment

DrawQRCode

Draw QR code

Syntax

```
public boolean DrawQRCode(String pszString, int x, int y, int nQRCodeUnitWidth, int nVersion, int nEcLevel)
```

Parameters

pszString

QR code text

x

It assigns the dots from the start point position to printing area margin in horizontal direction.(Horizontal coordinates)

It supports left justifying, mediate, right justifying

Introduce x as below chart

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

It assigns the dots from start point position to printing area margin in vertical direction(horizontal coordinates)

nQRCodeUnitWidth

QR code unit width, the range is [1,16].

QR code unit is wider, QR code is bigger.

nVersion

QR code version. 0 means to calculate version automatically.

QR code version is bigger, the edited characters are more, and the QR code is bigger.

nEcLevel

QR code error correction level.[1,4]

Return value

If the write command successfully, return TURE, or return FALSE

Remarks

DrawBitmap

Draw BMP bitmap

Syntax

```
public boolean DrawBitmap(Bitmap mBitmap, int x, int y, int dwWidth, int dwHeight, int nBinaryAlgorithm)
```

Parameters

mBitmap

src bitmap

x

It assigns the dots from the start point position to printing area margin in horizontal direction.(Horizontal coordinates)

It only supports left justifying, mediate, right justifying

Introduce x as below chart

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

It assigns the dots from start point position to printing area margin in vertical direction(vertical coordinates)

dwWidth

The printed width

dwHeight

The printed height

nBinaryAlgorithm

algorithm used to convert a color bitmap to binary bitmap

0 Use dithering algorithm for color images have better effect

1 Using the average threshold algorithm, and has better effect on text

class picture

Return value

If the write command successfully, return TURE, or return FALSE

Remarks

Canvas

Canvas communicate with printer via a IO object

Use Set(IO) to set Canvas IO object

The subsequent commands are transmitted via assigned IO

Set

Assign IO object

Syntax

```
public void Set(IO io)
```

Parameters

io

Need to use IO object

Return value

Remarks

Call this function to bind the bottom read and write object to the top logic object.

GetIO

Get the current IO object

Syntax

```
public IO GetIO() ()
```

Parameters

Return value

Current IO object

Remarks

CanvasBegin

Initialize the specified high wide canvas Subsequent drawing are within the scope of the canvas

Syntax

```
public void CanvasBegin(int width, int height)
```

Parameters

width

Canvas width (pixels, no more than a printer can print 58 mm printers use 384 mm width suggested printers use 576)

height

Canvas height

Return value

Remarks

CanvasEnd

After the end of the canvas, only can call the CanvasPrint for printing operations Do not do other operations

Syntax

```
public void CanvasEnd()
```

Parameters

Return value

Remarks

CanvasPrint

Print canvas content

Syntax

```
public void CanvasPrint(int nBinaryAlgorithm, int nCompressMethod)
```

Parameters

nBinaryAlgorithm

algorithm used to convert a color bitmap to binary bitmap

0 Use dithering algorithm for color images have better effect

1 Using the average threshold algorithm, and has better effect on text

class picture

nCompressMethod

use compress method

0 don't use compress method

1 use compress method

Return value

Remarks

SetPrintDirection

set print direction

Syntax

```
public void SetPrintDirection(int direction)
```

Parameters

`direction`

- 0 from left to right
- 1 from bottom to top
- 2 from right to left
- 3 from top to bottom

Return value

Remarks

DrawText

Draw text

Syntax

```
public void DrawText(String text, float x, float y, float rotation,
Typeface typeface, float textSize, int nFontStyle)
```

Parameters

text

text content

x

specify number of dots from starting area of horizontal direction to left margin of printing area. (horizontal axis)

support left center right alignment

incoming x see table

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

specify number of dots from starting area of vertical direction to upper edge of printing area.(vertical axis)

support top center bottom alignment

incoming x see table

y	implication
-1	top alignment
-2	center alignment
-3	right alignment
Great than or equal to 0	vertical direction

rotation

rotate digree (clockwise)

typeface

font type

textSize

font size

nFontStyle

```
font style
    0x00 normal
    0x08 bold
    0x80 underline
```

Return value

Remarks

DrawTextAutoNewLine

Draw text

Syntax

```
public void DrawTextAutoNewLine(String text, float x, float y, float rotation, Typeface typeface, float textSize, int nFontStyle, int outerwidth, float spacingmult, float spacingadd)
```

Parameters

text

x

y

rotation

typeface

textSize

nFontStyle

See DrawText

outerwidth

More than that width automatically breaks

spacingmult

The relative font height line spacing

spacingadd

Additional line spacing added

Return value

Remarks

DrawTextMultiLine

Draw text

Syntax

```
public void DrawTextMultiLine(String text, float x, float y, float rotation, Typeface typeface, float textSize, int nFontStyle, float rowWidth, float lineSpacing)
```

Parameters

Return value

Remarks

Equal to DrawTextAutoNewLine, Please use DrawTextAutoNewLine

DrawLine

draw a line

Syntax

```
public void DrawLine(float startX, float startY, float stopX, float stopY)
```

Parameters

`startX`

start point horizontal position

`startY`

start point vertical position

`stopX`

end point horizontal position

`stopY`

end point vertical position

Return value

Remarks

DrawBox

draw a box

Syntax

```
public void DrawBox(float left, float top, float right, float bottom)
```

Parameters

left

x axis left upper corner of the box

top

y axis left upper corner of the box

right

x axis right bottom corner of the box

bottom

y axis right bottom corner of the box

Return value

Remarks

DrawRect

draw a rect

Syntax

```
public void DrawRect(float left, float top, float right, float bottom)
```

Parameters

`left`

x axis left upper corner of the rectangle

`top`

y axis left upper corner of the rectangle

`right`

x axis right bottom corner of the rectangle

`bottom`

y axis right bottom corner of the rectangle

Return value

Remarks

DrawBitmap

draw bitmap

Syntax

```
public void DrawBitmap(Bitmap bitmap, float x, float y, float rotation)
```

Parameters

bitmap

bitmap

x

specify number of dots from starting area of horizontal direction to left margin of printing area. (horizontal axis)

support left center right alignment

incoming x see table

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

specify number of dots from starting area of vertical direction to upper edge of printing area.(vertical axis)

support top center bottom alignment

incoming x see table

y	implication
-1	top alignment
-2	center alignment
-3	right alignment
Great than or equal to 0	vertical direction

rotation

rotate degree (clockwise)

Return value

Remarks

DrawQRCode

draw qr code

Syntax

```
public void DrawQRCode(String text, float x, float y, float rotation, int  
unitWidth, int version, int ecc)
```

Parameters

text

qr code content

x

specify number of dots from starting area of horizontal direction to left margin of printing area. (horizontal axis)

support left center right alignment

incoming x see table

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

specify number of dots from starting area of vertical direction to upper edge of printing area.(vertical axis)

support top center bottom alignment

incoming x see table

y	implication
-1	top alignment
-2	center alignment
-3	right alignment
Great than or equal to 0	vertical direction

rotation

rotate degree (clockwise)

unitWidth

QR code unit width, the range is [1,16].

QR code unit is wider, QR code is bigger.

version

QR code version. 0 means to calculate version automatically.

QR code version is bigger, the edited characters are more, and the QR code is bigger.

`ecc`

QR code error correction level.[1,4]

Return value

Remarks

DrawBarcode

draw barcode

Syntax

```
public void DrawBarcode(String text, float x, float y, float rotation,
int unitWidth, int height, int barcodeType)
```

Parameters

text

barcode content

x

specify number of dots from starting area of horizontal direction to left margin of printing area. (horizontal axis)

support left center right alignment

incoming x see table

x	implication
-1	Left alignment
-2	Center alignment
-3	Right alignment
Great than or equal to 0	Horizontal direction

y

specify number of dots from starting area of vertical direction to upper edge of printing area.(vertical axis)

support top center bottom alignment

incoming x see table

y	implication
-1	top alignment
-2	center alignment
-3	right alignment
Great than or equal to 0	vertical direction

rotation

rotate degree (clockwise)

unitWidth

barcode unit width, range[1,16]. the bigger barcode, the larger unitWidth

height

barcode height(normally in dpi 203, 8 point means 1mm)
set height to 80, the barcode printed is 1CM.

barcodeType

currently only support CODE128

incoming value see table

Value	Meaning
0x49	CODE 128

Return value

Remarks